

HDOC DAY -16

Write a menu-driven C program using switch-case to perform digit-based operations on a positive integer.

Define one separate user-defined function for each menu option, with void return type and a simple function call (without call by value/reference). Each function should read the required input, process it, and display the result.

The menu options are:

1. Check whether its digits are in increasing order from left to right.
2. Check whether its digits are in decreasing order from left to right.
3. Find the largest digit and smallest digit.
4. Find the second largest digit and second smallest digit.

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared test cases.

Aim:

To write a menu-driven C program using switch-case and user defined function to perform different digit-based operations on a positive integer.

Algorithm

1. **Start**
2. Display the menu:
 - 1 → Check increasing order
 - 2 → Check decreasing order
 - 3 → Find largest and smallest digit
 - 4 → Find second largest and second smallest digit
3. Enter the user's **choice**.
4. Use **switch-case**.
5. According to the choice, call the respective function.
6. Inside the function, input a **positive integer**.
7. Perform the required digit operation.
8. Display the result.
9. **Stop**.

Test Case	Choice	Input Number	Expected Output
1	1	12345	Digits are in increasing order
2	2	98721	Digits are in decreasing order
3	3	58321	Largest = 8, Smallest = 1
4	4	58321	Second Largest = 5, Second Smallest = 2

>Write a code c.

```

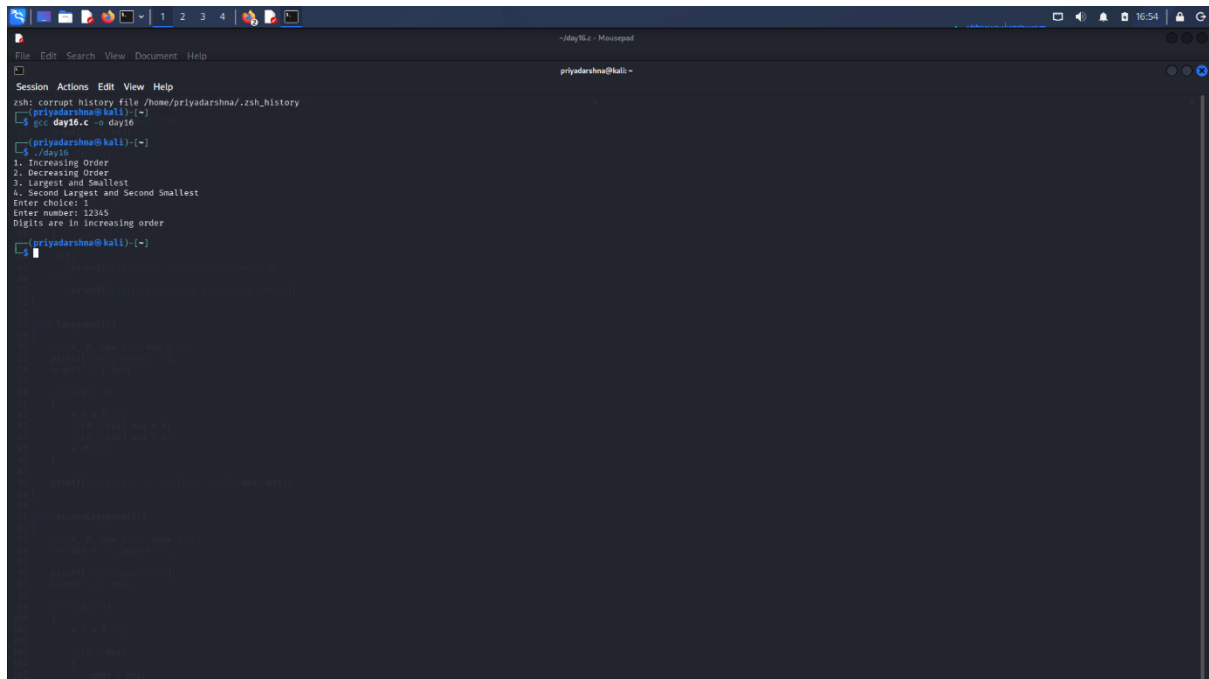
1 #include <stdio.h>
2
3 void increasing();
4 void decreasing();
5 void largeSmall();
6 void secondLargeSmall();
7
8 int main()
9 {
10     int ch;
11
12     printf("1. Increasing Order\n");
13     printf("2. Decreasing Order\n");
14     printf("3. Largest and Smallest\n");
15     printf("4. Second Largest and Second Smallest\n");
16
17     printf("Enter choice: ");
18     scanf("%d", &ch);
19
20     switch(ch)
21     {
22         case 1: increasing(); break;
23         case 2: decreasing(); break;
24         case 3: largeSmall(); break;
25         case 4: secondLargeSmall(); break;
26         default: printf("Invalid choice");
27     }
28
29     return 0;
30 }
31
32 void increasing()
33 {
34     int n, d, p = 10, f = 1;
35     printf("Enter number: ");
36     scanf("%d", &n);
37
38     while(n > 0)
39     {
40         d = n % 10;
41         if(d > p)
42             f = 0;
43         p = d;
44         n /= 10;
45     }
46
47     if(f)
48         printf("Digits are in increasing order");
49     else
50         printf("Digits are not in increasing order");

```

```
51 }
52
53 void decreasing()
54 {
55     int n, d, p = -1, f = 1;
56     printf("Enter number: ");
57     scanf("%d", &n);
58
59     while(n > 0)
60     {
61         d = n % 10;
62         if(d < p)
63             f = 0;
64         p = d;
65         n /= 10;
66     }
67
68     if(f)
69         printf("Digits are in decreasing order");
70     else
71         printf("Digits are not in decreasing order");
72 }
73
74 void largestSmall()
75 {
76     int n, d, max = 0, min = 9;
77     printf("Enter number: ");
78     scanf("%d", &n);
79
80     while(n > 0)
81     {
82         d = n % 10;
83         if(d > max) max = d;
84         if(d < min) min = d;
85         n /= 10;
86     }
87
88     printf("Largest = %d\nSmallest = %d", max, min);
89 }
90
91 void secondLargestSmall()
92 {
93     int n, d, max = -1, smax = -1;
94     int min = 10, smin = 10;
95
96     printf("Enter number: ");
97     scanf("%d", &n);
98
99     while(n > 0)
100     {
```

```
100     {
101         d = n % 10;
102
103         if(d > max)
104         {
105             smax = max;
106             max = d;
107         }
108         else if(d > smax && d != max)
109             smax = d;
110
111         if(d < min)
112         {
113             smin = min;
114             min = d;
115         }
116         else if(d < smin && d != min)
117             smin = d;
118
119         n /= 10;
120     }
121
122     printf("Second Largest = %d\n", smax);
123     printf("Second Smallest = %d", smin);
124 }
125 }
```

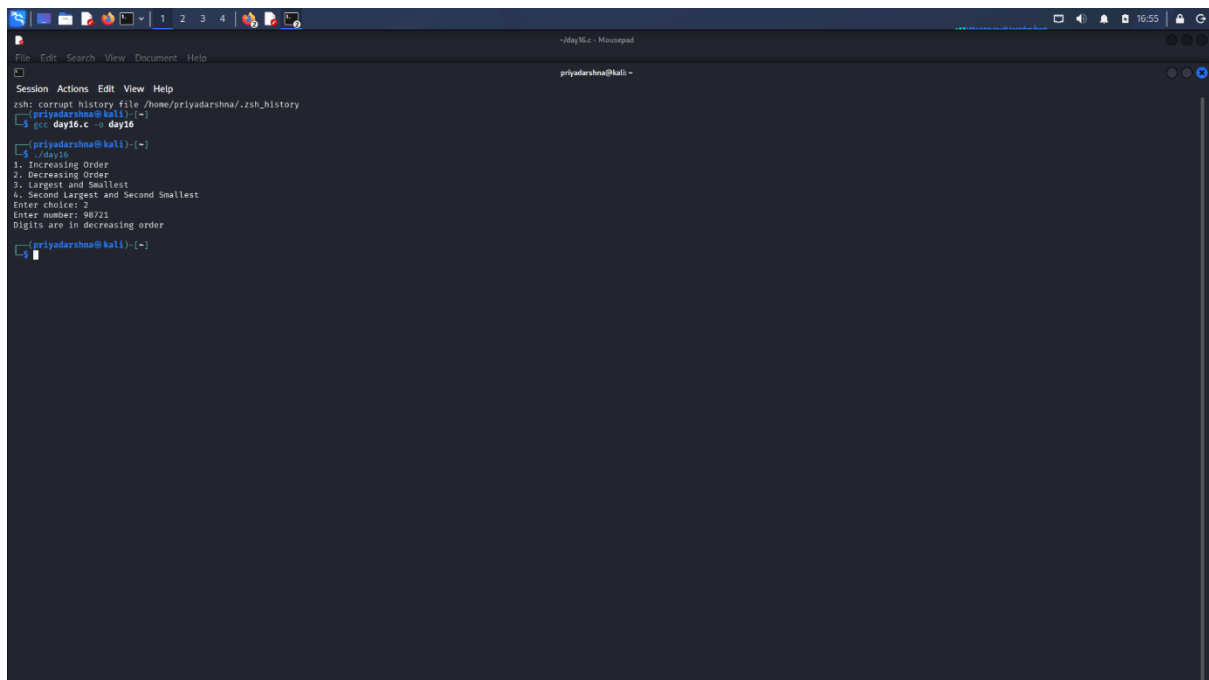
>Test Case 1;



The screenshot shows a terminal window with a dark background. The prompt is `priyadarshna@kali:~`. The user has run `gcc day16.c -o day16` and then `./day16`. The program displays a menu with four options: 1. Increasing Order, 2. Decreasing Order, 3. Largest and Smallest, and 4. Second Largest and Second Smallest. The user has entered choice 1. The program then asks for a number, and the user has entered 12345. The output is "Digits are in increasing order".

```
priyadarshna@kali:~  
$ gcc day16.c -o day16  
priyadarshna@kali:~  
$ ./day16  
1. Increasing Order  
2. Decreasing Order  
3. Largest and Smallest  
4. Second Largest and Second Smallest  
Enter choice: 1  
Enter number: 12345  
Digits are in increasing order  
priyadarshna@kali:~  
$
```

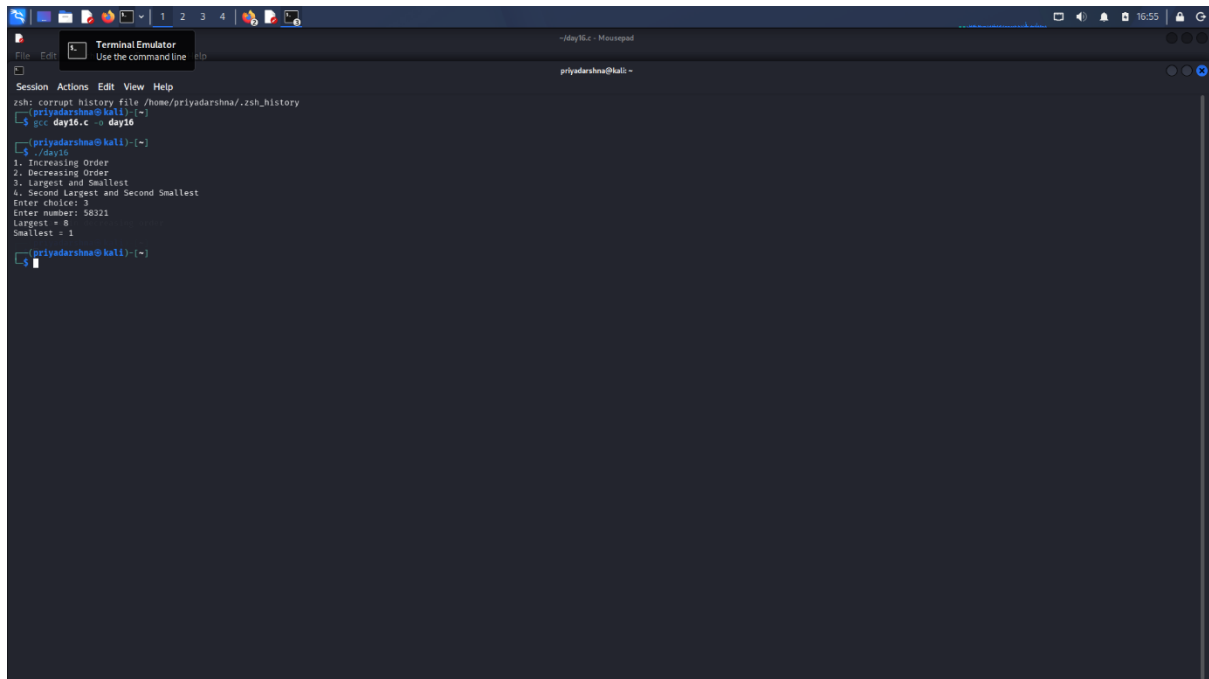
>Test Case 2;



The screenshot shows a terminal window with a dark background. The prompt is `priyadarshna@kali:~`. The user has run `gcc day16.c -o day16` and then `./day16`. The program displays the same menu as in Test Case 1. The user has entered choice 2. The program then asks for a number, and the user has entered 98721. The output is "Digits are in decreasing order".

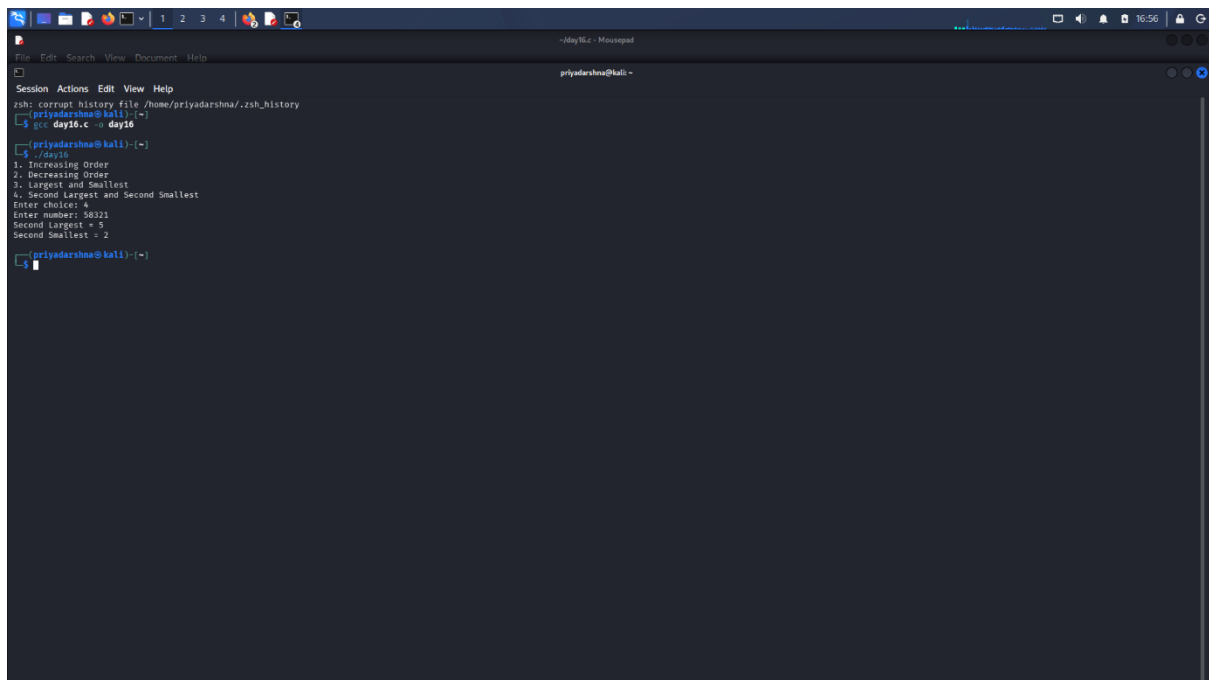
```
priyadarshna@kali:~  
$ gcc day16.c -o day16  
priyadarshna@kali:~  
$ ./day16  
1. Increasing Order  
2. Decreasing Order  
3. Largest and Smallest  
4. Second Largest and Second Smallest  
Enter choice: 2  
Enter number: 98721  
Digits are in decreasing order  
priyadarshna@kali:~  
$
```

>Test Case 3;



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
~ (priyadarshna@kali) [-]
$ gcc day16.c -o day16
~ (priyadarshna@kali) [-]
$ ./day16
1. Increasing Order
2. Decreasing Order
3. Largest and Smallest
4. Second Largest and Second Smallest
Enter choice: 3
Enter number: 58321
Largest = 8
Smallest = 1
~ (priyadarshna@kali) [-]
```

>Test Case 4;



```
zsh: corrupt history file /home/priyadarshna/.zsh_history
~ (priyadarshna@kali) [-]
$ gcc day16.c -o day16
~ (priyadarshna@kali) [-]
$ ./day16
1. Increasing Order
2. Decreasing Order
3. Largest and Smallest
4. Second Largest and Second Smallest
Enter choice: 4
Enter number: 58321
Second Largest = 5
Second Smallest = 2
~ (priyadarshna@kali) [-]
```

Result:

The menu-driven C program was compiled and executed successfully , and the required digit operations were performed correctly.